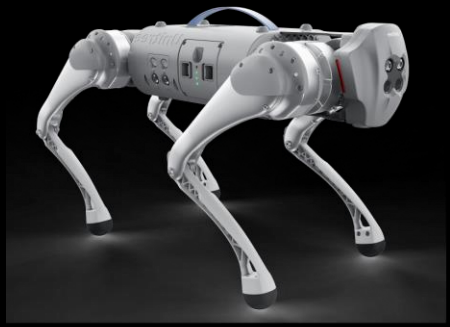
Weston Robot

Basic Training

Go1: Robot Dog



Confidential

www.WestonRobot.com

Agenda

01
Main Components

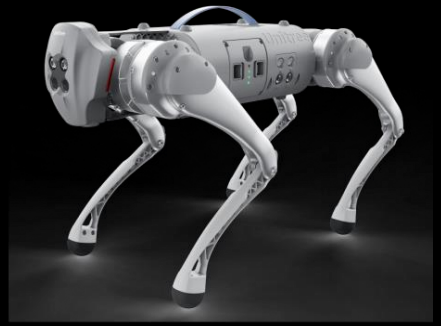
02
Basic Operations

03
Development

Main Components

Weston Robot

Main Components



Go1

Battery

Controller

Battery Charger

Label Controller

Emergency Stop

Weston Robot

Basic Operations

Switching On the Robot

- How to
  - Insert the battery
  - Do a short press followed by a long press on the On button
- Safety Precaution
  - Place Go1 in Neutral Prone Position
- Battery Precaution
  - Ensure battery level > 1 LED
  - Ensure battery was not just charged (Heating)

Switching On the Robot

- Battery
  - Battery Level Display
  - Use provided charger and power supply
  - Overcharge Protection
  - Self-discharge Protection (65%)

  Note: Do not charge battery without supervision

- Controller
  - Charge using USB – C cable provided
  - Tap then Hold the On button (Beep + Lights)
  - Tapping every button causes a Beep

Remote Controller Commands

Button		Effect
Left Stick	Back and Forth	Move back and forth (body height up and down)
	Left and Right	Lateral movement/(tilt head)
Right Stick	Back and Forth	Head up/down (pitch)
	Left and Right	Twist/(rotation)
SELECT		Standing (Mode 1)
START		Unlock the joints
		Move (Mode 2)
		Constantly moving (Double tap to enter Mode 3)
L2 + START (Press together)		Running mode
L1 + A (Press together)		Praying
L1 + X (Press together)		Dance 1
R1 + X (Press together)		Dance 2
L2 + X (Press together)		Recover
L2 + Y (Press together)		Backflip
R2 + LEFT (Press together)		Jump to left
R2 + RIGHT (Press together)		Jump to right
Moving mode (Mode 2) + A		Stairs climbing mode
Stairs climbing mode + B		Resume moving mode
Posture Switching		
L2 + B		Damping State
L2 + A		Lock the joints and enter prone position
L2 + A		Enter standing posture (joint locking)

Note: Do not use these commands for Go1 with LiDAR attached

Adjust Body Height

- Instructions
  - In constantly Walking Mode
  - *Press Up / Down Arrows*

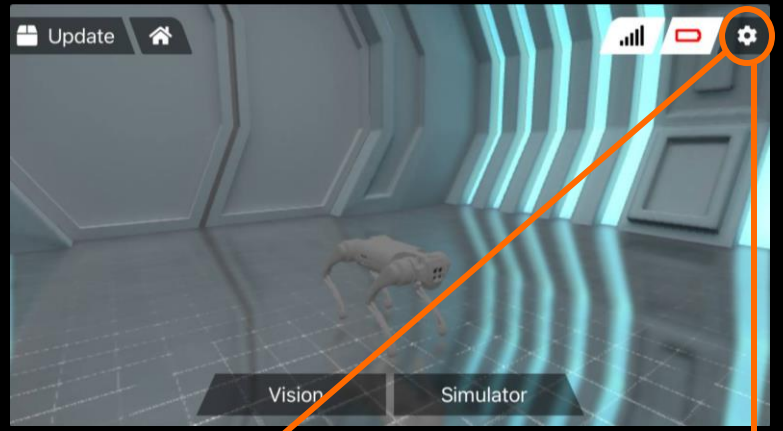
- Precaution
  - Remember number of key presses
  - Return to default setting to conserve battery
  - Carefully control speed
  - ONLY do this in walking mode

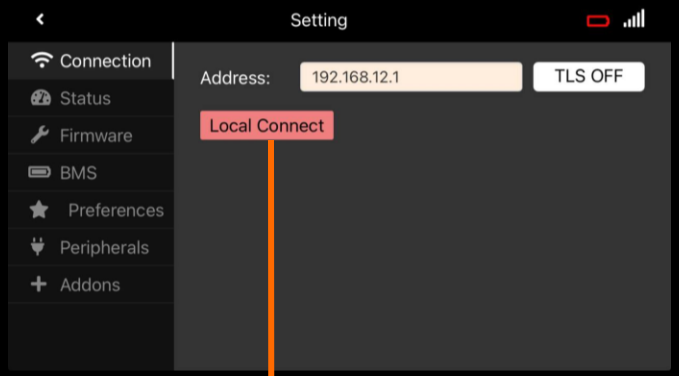
Basic App Usage (beta)

1. Download Go1 App
   - https://www.unitree.com/app/
   - iOS
     - https://testflight.apple.com/join/KraKgqam
   - Android
     - https://unitreeapp.oss-cn-beijing.aliyuncs.com/Go1_App.zip

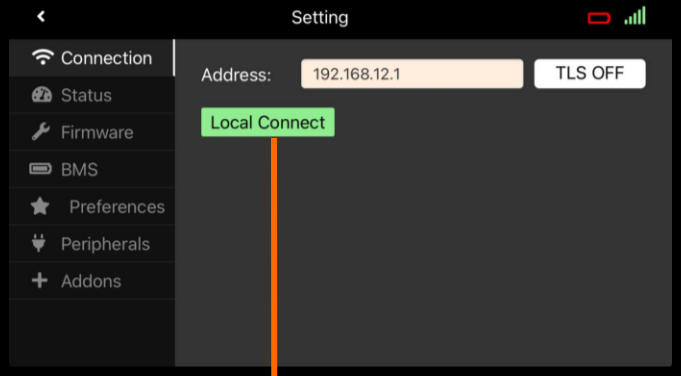
1. Connect to Go1: WiFi
   - Unitree_Go1XXXXXX
   - Password: 00000000 (Eights Zeros)
   - Ip Setting: DHCP

Go1 App: Check Connectivity



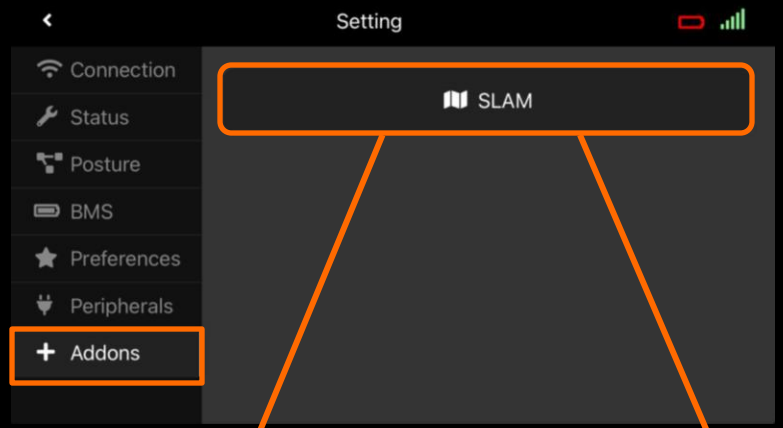


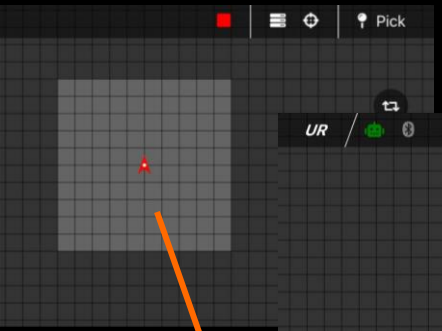
Disconnected

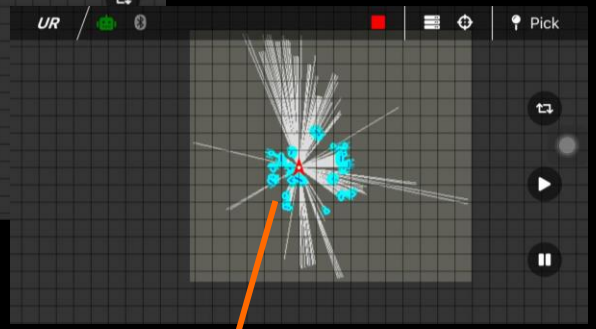


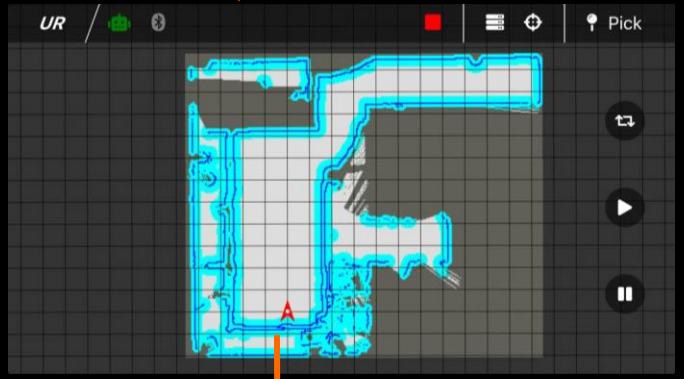
Connected

Go1 App: Check Map





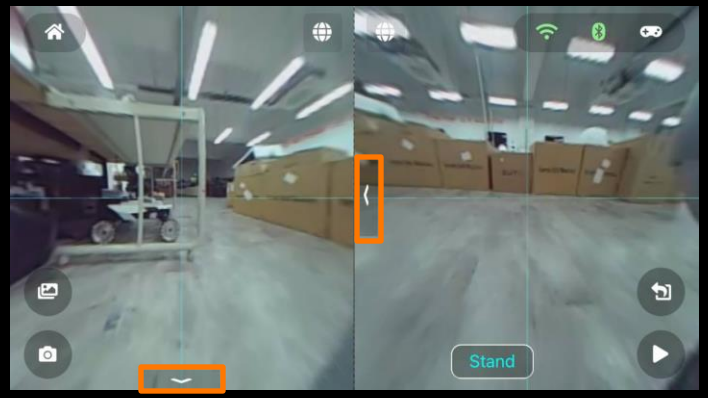


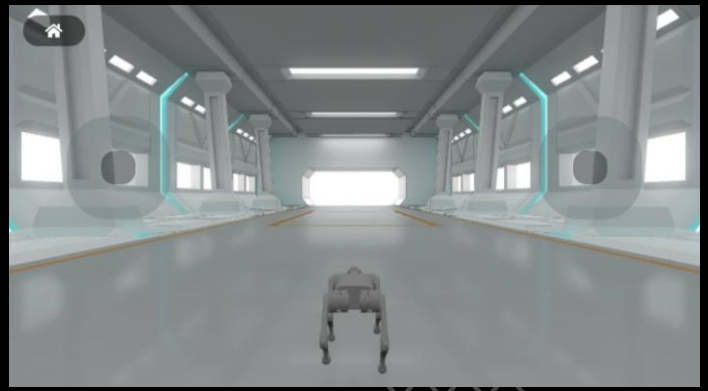


No Map Loaded

Map Loaded Correctly

Go1 App





Stairs Climbing Mode

Note: Due to the limited leg length, there is restriction on the height of climbing stairs of < 12cm

Usable Actions for Go1 (with LiDAR)

Note: Do not use actions highlighted in Red for Go1 with LiDAR attached

Follow Me Tag

1. Preparation: Wear the tag on the right waist, stand on the left side of Go1, and keep a distance of about 1 meter
2. Power-on pairing: Short press followed by a long press on the POW button
3. Start: Double-click on the MOD button
4. Run: Double-click on the MOD button again
5. Stop and stand: Click the MOD button or push the joystick
6. Switch Go1 posture: double-click on the POW button

Do NOT operate Go1 when

- Slope > 22°
- Slippery / Soft Floor
- Low Battery Levels
- Use Other batteries

Development

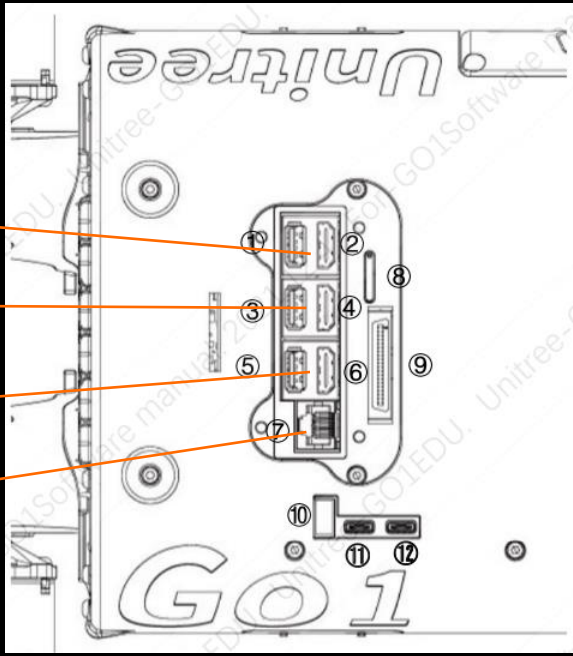
Internal Components of Go1

Nano

NX

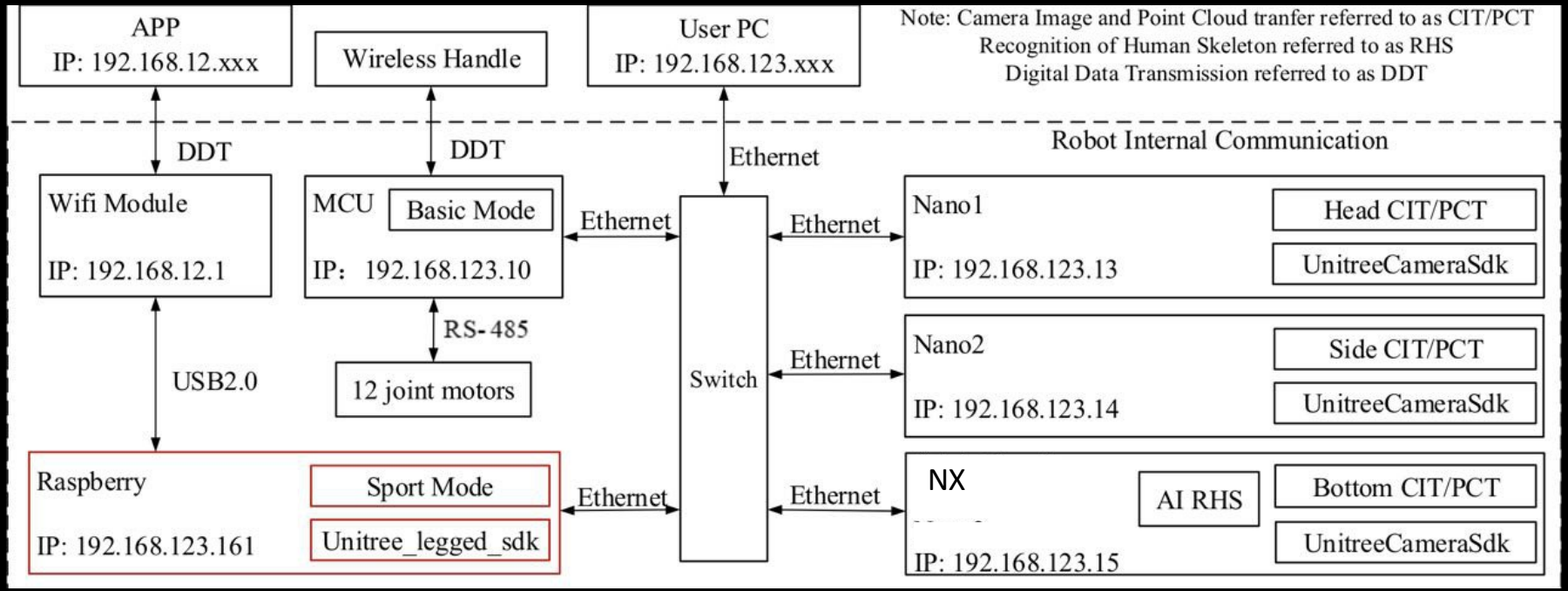
Raspberry Pi (RPi)

Ethernet

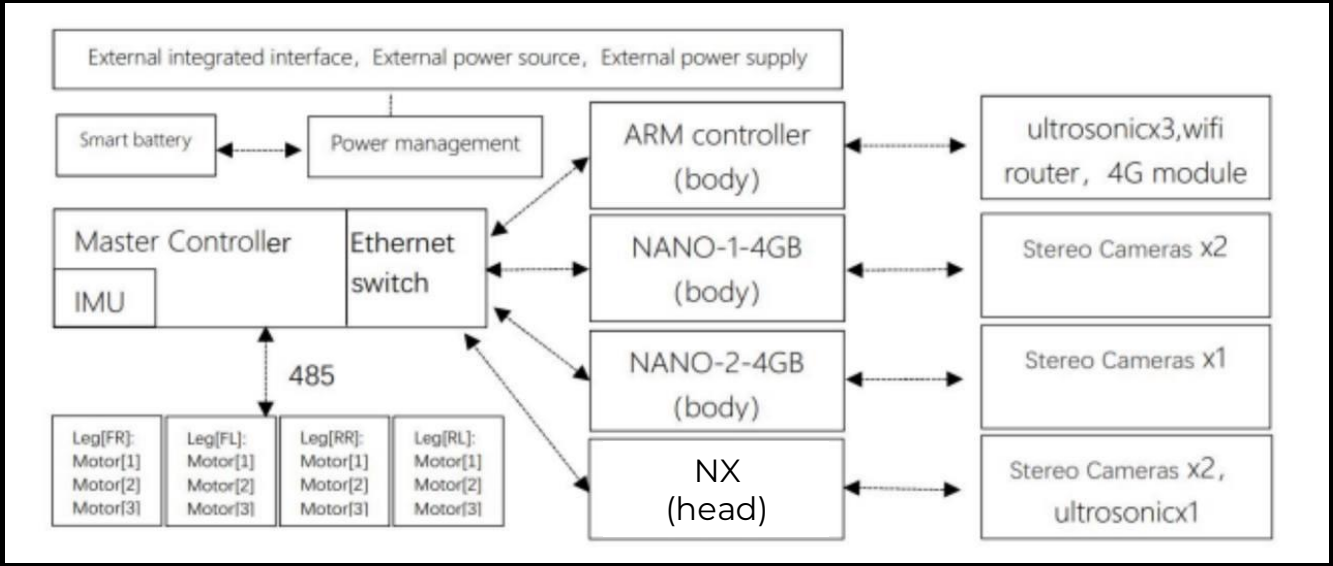


1. Nano2 USB
2. Nano2 HDMI
3. Nano/Nx3 USB
4. Nano/Nx3 HDMI
5. Raspberry USB
6. Raspberry HDMI
7. Gigabit Ethernet port
8. SIM
9. Integratedinterface
10. External power input (24V 30A)
11. Type-C
12. Type-C

System Architecture Diagram



System Architecture Diagram



Access RPi / Nano1 / Nano2 / NX

- Hardware
  - HDMI Monitor
  - Mouse
  - Keyboard

- Network (192.168.123.X)
  - 192.168.123.161 (RPi)
  - 192.168.123.13 (Nano1)
  - 192.168.123.14 (Nano2)
  - 192.168.123.15 (NX)

Connect to RPi

- Wi-Fi
  - Unitree_Go1XXXXXX
  - Password: 00000000

- Log in
  - IP Address: 192.168.123.161 (External Network)
  - Username: pi
  - Password: 123

- Example:
  - ssh pi@192.168.123.161
  - Password: 123

Connect to Nano1/Nano2/ NX

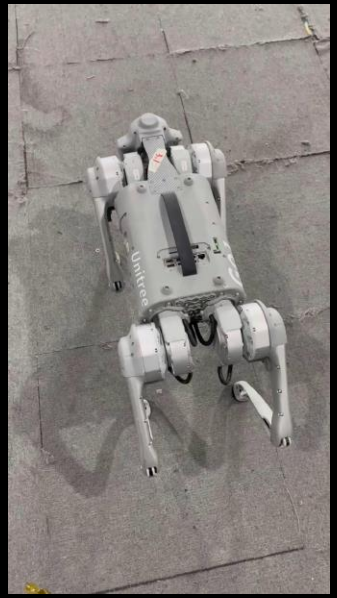
- SSH into RPi
  - ssh pi@192.168.123.161
  - Password: 123

- Log in
  - IP Address: 192.168.123.13/14/15 (External Network)
  - Username: unitree
  - Password: 123

- Example:
  - ssh unitree@192.168.123.13/14/15
  - Password: 123

unitree_legged_sdk

- Unitree provides high-level as well as low-level control routines as shown in the video.
- The high-level is the robot as the control object, giving instructions related to travel speed, etc. No need to build complex dynamics controllers
- The low-level is the motors as the control object, which controls the robot through three loops: speed, position, and torque.



High-Level

Low-Level

unitree_legged_sdk

- SSH into RPi
  - ssh pi@192.168.123.161
  - Password: 123
- unitree_legged_sdk examples (High–Level)
  - cd Unitree/sdk/unitree_legged_sdk/
  - mkdir build && cd build
  - cmake ../
  - make
  - sudo ./example_walk

Note: All examples have to be run by sudo command

example_walk.cpp	HighState add MotorState	high level control

unitree_legged_sdk

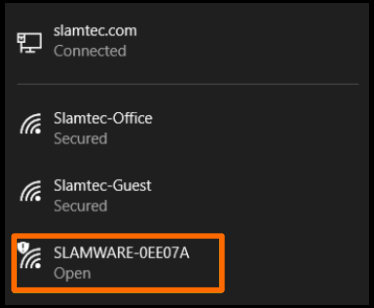
- SSH into RPi
  - ssh pi@192.168.123.161
  - Password: 123
- unitree_legged_sdk examples (Low–Level)
  - Hang Go1 in a safe position
  - L2+B(Together) - robot limbs will drop to ground
  - L1+L2+START(Together) - robot gets into normal mode
  - sudo ./example_velocity
  - Exit normal mode: Restart Go1 OR cd Unitree/autostart/sportsMode
    - sudo ./bin/Legged_sport

Note: All examples have to be run by sudo command

example_position.cpp	version 3.4	low level-position control
example_torque.cpp	version 3.4	low level-torque control
example_velocity.cpp	version 3.4	low level-velocity control

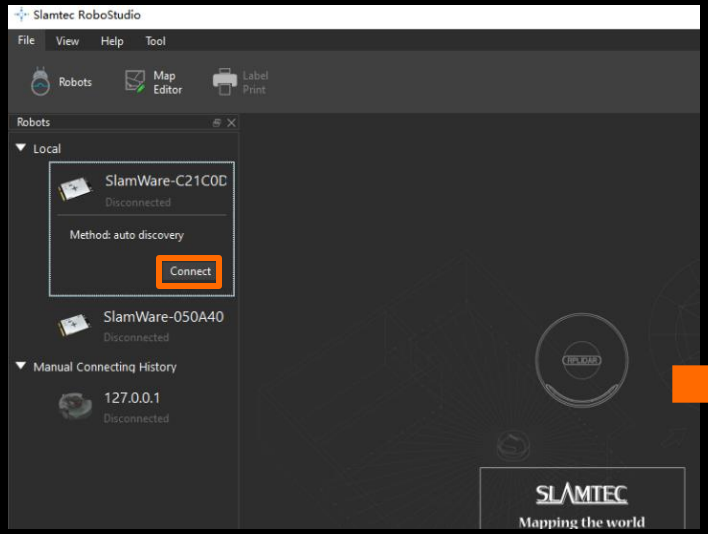
RoboStudio (2D LiDAR)

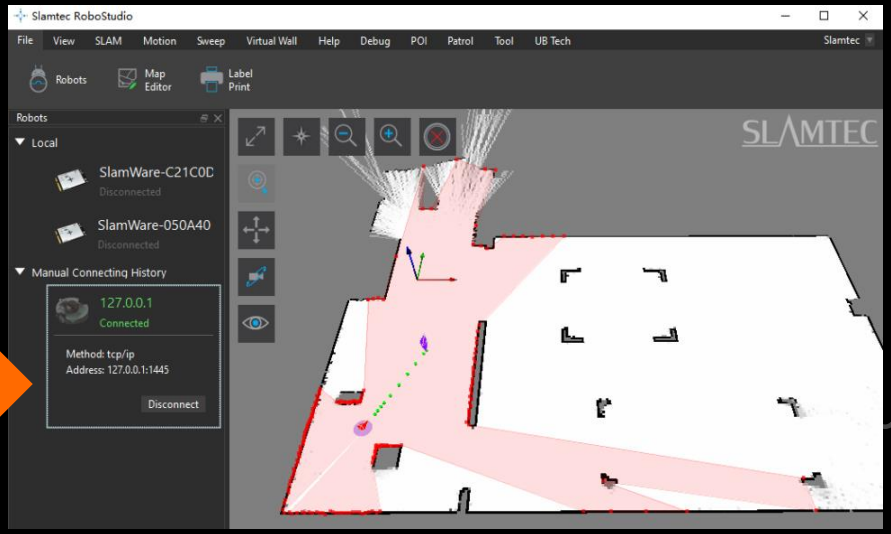
- Install RoboStudio on Windows/Android devices
  - https://www.slamtec.com/en/RoboStudio
- SLAMWARE User Manual
  - https://bucket-download.slamtec.com/e16008e591546788a026fb8c21204088ed08e24d/SA008_SLAMTEC_slamware_robot_studio_appnote_v1.1_en.pdf
- Connect to SLAMWARE Wi-Fi



2D Mapping

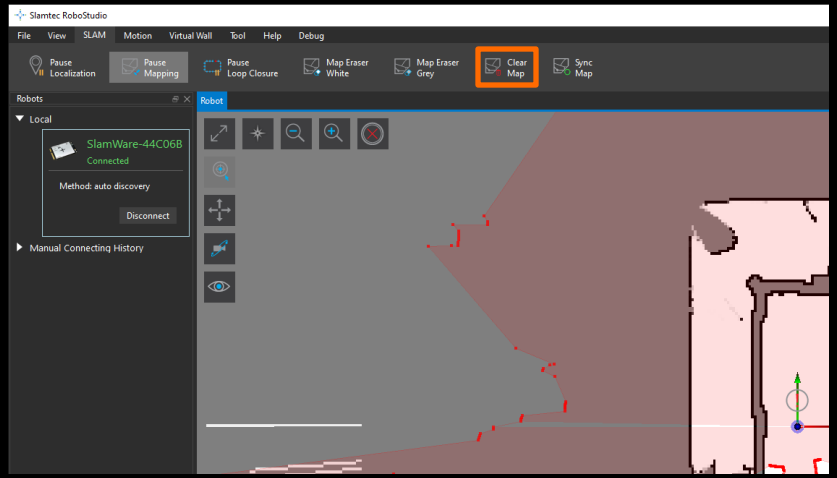
- Connect Robot





2D Mapping

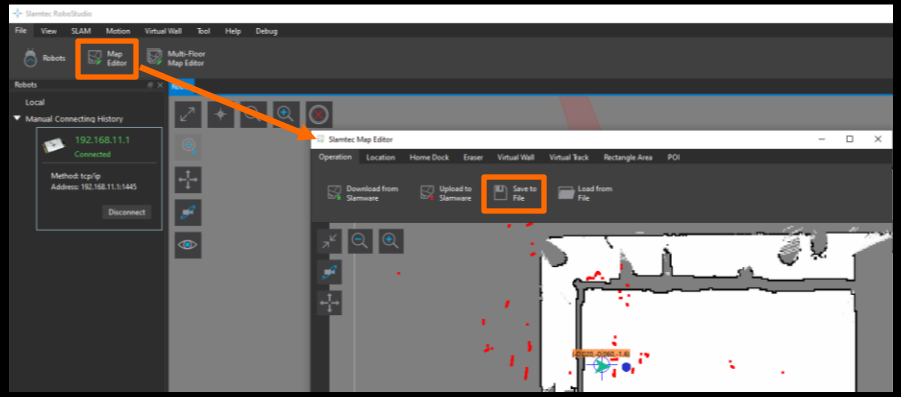
- Clear current map and start mapping



- When mapping, drive Go1 in low speed and **turn extremely slowly**, otherwise robot will be delocalized which will result in inaccurate map

2D Mapping

- When mapping is done, Save map by clicking on the Save to File button



- Name map as map.stcm and transfer new map file to NX board and place new map under this directory:
  /home/unitree/catkin_ws/src/slamware_ros_sample/map
  OR
  /home/Unitree/UnitreeSLAM/catkin_lidar_slam_2d_go1/src/slamrplidarV2.0/slam_planner/maps

Ultrasonic Sensor Readings

- Log into RPi
  - Go to folder: Unitree>sdk>ultrasonic_example
  - You can compile this `CMake` project as follows.
    - mkdir build && cd build
    - cmake ..
    - make
  - You can run the listener like this:
    - cd build
    - ./ultrasonic_listener_lcm

Camera Data

- Dependencies*
  - OpenCV (version 4.11)
  - CMake (version 2.8 or higher)
  - OpenGL
  - GLUT
  - X11

*Note: If you are using the onboard computer of Go1, you can skip the step of installing dependencies

Camera Data

- Log into Nano/NX
- Enter top command in terminal

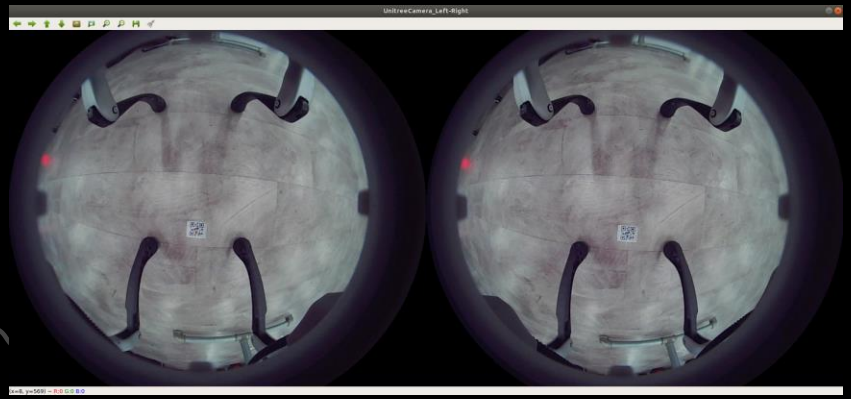
```
unitree@unitree-desktop:~$ top
```

- Kill processes "point_cloud_nod"

```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU %MEM     TIME+ COMMAND
 7710 root      20   0  662316  87376  60292 S 127.5  2.2   2:05.17 point_cloud_nod
 7735 root      20   0  660244  87344  60496 S 120.7  2.2   1:54.80 point_cloud_nod
```

Camera Data

- Go to folder: Unitree>sdk>UnitreeCameraSdk
- You can compile this `CMake` project as follows.
  - mkdir build && cd build
  - cmake ..
  - make
- You can run the files like this:
  - ./bins/example_getRawFrame



Camera Data

- Log into Nano/NX
  - Go to folder: Unitree>sdk>UnitreeCameraSdk
  - You can compile this `CMake` project as follows.
    - mkdir build && cd build
    - cmake ..
    - make
  - You can run the files with sudo permission like this:
    - sudo ./bins/example_getRawFrame

example_getCalibParamsFile.cc	Get the internal parameters of the camera
example_getDepthFrame.cc	Get camera depth video stream
example_getPointCloud.cc	Get camera point cloud
example_getRawFrame.cc	Get camera RGB binocular video stream
example_getRectFrame.cc	Obtain RGB binocular video stream after camera correction